

Code-Layout, Readability and Reusability

Date	14 July 2026
Team ID	
Project Name	SMART LENDER
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout	Yes	PEP 8 spacing followed in Python and Flask scripts.
2	Proper File Structure	Files and folders are logically organized	Yes	Separate directories for templates/ and models/.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Descriptive names like applicant_income and credit_history.
4	Function / Method Names	Functions are descriptively named	Yes	Functions named decisionTree() and xgboost().
5	Code Comments	Inline and block comments explain logic	Yes	Inline comments explain SMOTE and model evaluation.
6	Modular Design	Code is split into reusable functions/modules	Yes	Created reusable utility functions for data scaling.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Removed redundant variables and unused imports.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Try-except blocks handle missing fields and model loading.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Data Scaling Utility	Python / Scikit-learn	Model Training & App Inference	High

2	Model Evaluation Module	Python / Scikit-learn	Testing Pipeline	High
3	JSON Helper	Python / Re	API response parser	Medium
4	UI input validator	JavaScript	predict.html Form	Medium
5				

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Clean and follows PEP 8 guidelines.
Readability	5	Descriptive naming conventions are used.
Reusability	4	Modular components can be reused.
Documentation / Comments	4	Well-commented code blocks.
Overall Score	4.5	Highly maintainable and clean code.